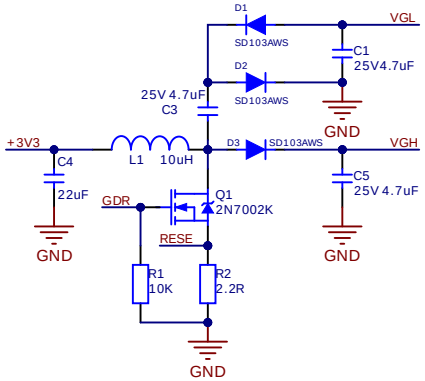




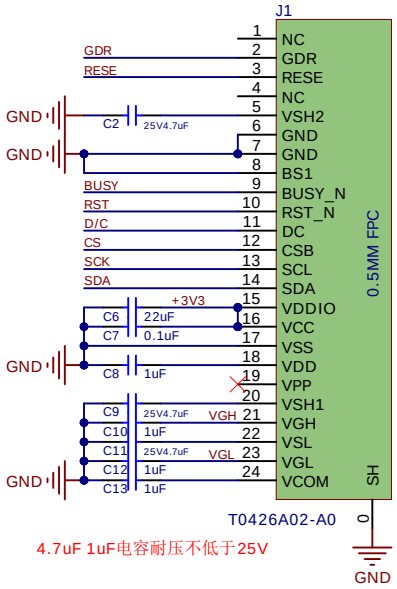


Table 5-1: Power Supply Pins

Name	Type	Connect to	Function	Description	When not in use
VCI	P	Power Supply	Power Supply	This pin is Power input pin for the chip.	-
VCIGS	P	Power Supply	Power Supply	This pin is Power input pin for the chip. - Connect to VCI in the application circuit.	-
VCIQS	P	Power Supply	Power Supply	This pin is Power input pin for the chip. - Connect to VCI in the application circuit.	-
VDDIO	P	Power Supply	Power for Interface logic pins	This pin is Power input pin for the Interface. - Connect to VCI in the application circuit.	-
VDD	P	Capacitor	Regulator output	This pin is Core logic power pin VDD can be regulated internally from VCI. A capacitor should be connected between VDD and VSS under all circumstances.	-
EXTVDD	I	VSS	Regulator bypass	This pin is VDD regulator bypass pin. EXTVDD should be connected to VSS in the application circuit.	-
VSS	P	VSS	GND	This pin is Ground pin	-
VSSGS	P	VSS	GND	This pin is Ground pin. - Connect to VSS in the application circuit.	-
VSSQS	P	VSS	GND	This pin is Ground pin. - Connect to VSS in the application circuit.	-
VPP	P	Power Supply	OTP power	This pin is Power Supply for OTP Programming.	Open

Table 5-2: Interface Logic Pins

Name	Type	Connect to	Function	Description	When not in use
SCL	I	MPU	Data Bus	This pin is Serial clock pin for interface: Refer to Session 6.1 - MCU Interface.	-
SDI	I	MPU	Data Bus	This pin is Serial data pin for interface: Refer to Session 6.1 - MCU Interface.	-
SDO	O				Open
CS#	I	MPU	Logic Control	This pin is the chip select input connecting to the MCU. Refer to Session 6.1 - MCU Interface.	VDDIO or VSS
D/C#	I	MPU	Logic Control	This pin is Data/Command control pin connecting to the MCU. Refer to Session 6.1- MCU Interface.	VDDIO or VSS
RES#	I	MPU	System Reset	This pin is reset signal input. Active Low.	-
BUSY	O	MPU	Device Busy Signal	This pin is Busy state output pin When Busy is High, the operation of the chip should not be interrupted, and command should not be sent. For example,. The chip would put Busy pin High when - Outputting display form; or - Programming with OTP - Communicating with digital temperature sensor	Open
BS1	I	VDDIO/VSS	MCU Interface Mode Selection	This pin is for selecting 3-wire or 4-wire SPI bus. BS1 | MCU Interface L | 4-wire SPI H | 3-wire SPI(9 bits SPI)	-
M/S#	I	VDDIO	Reserved for Testing	This pin is reserved pin and should be connected to VDDIO.	-
CL	I/O	NC	Clock signal	This is the clock signal pin. It should be left open in application.	Open
TSDA	I/O	Temperature sensor SDA	Interface to Digital Temp. Sensor	This pin is I²C Interface to digital temperature sensor Data pin. External pull up resistor is required when connecting to I²C slave.	Open
TSCL	O	Temperature sensor SCL	Interface to Digital Temp. Sensor	This pin is I²C Interface to digital temperature sensor Clock pin. External pull up resistor is required when connecting to I²C slave.	Open

Table 5-3: Analog Pins

Name	Type	Connect to	Function	Description	When not in use
GDR	O	POWER MOSFET Driver Control	VGHR, VGLR Generation	This pin is N-Channel MOSFET gate drive control pin.	-
RESE	I	Booster Control Input		This pin is Current sense input pin for the control Loop.	-
VGHR	C	Stabilizing capacitor		This pin is Positive Gate driving voltage regulation. Connect a stabilizing capacitor between VGHR and VSS in the application circuit.	Open
VGLR	C	Stabilizing capacitor		This pin is Negative Gate driving voltage regulation. Connect a stabilizing capacitor between VGLR and VSS in the application circuit.	Open
VSH1R	C	Stabilizing capacitor	VSH1R, VSH2R, VSLR Generation	This pin is Positive Source driving voltage regulation - VSH1R. Connect a stabilizing capacitor between VSH1R and VSS in the application circuit.	Open
VSH2R	C	Stabilizing capacitor		This pin is Positive Source driving voltage regulation – VSH2R. Connect a stabilizing capacitor between VSH2R and VSS in the application circuit.	Open
VSLR	C	Stabilizing capacitor		This pin is Negative Source driving voltage regulation. Connect a stabilizing capacitor between VSLR and VSS in the application circuit.	Open
VGH	P	VGH supply		This pin is Positive Gate driving voltage.	VGHR
VGL	P	VGL supply		This pin is Negative Gate driving voltage.	VGLR
VSH1	P	VSH1 supply		This pin is Positive Source driving voltage - VSH1.	VSH1R
VSH2	P	VSH2 supply		This pin is Positive Source driving voltage - VSH2.	VSH2R
VSL	P	VSL supply		This pin is Negative Source driving voltage.	VSLR

Table 5-4: External Power Supply Pins

Name	Type	Connect to	Function	Description	When not in use
VGH	P	VGH supply	Analog Pins for External Power Supply	This pin is Positive Gate driving voltage.	VGHR
VGL	P	VGL supply		This pin is Negative Gate driving voltage.	VGLR
VSH1	P	VSH1 supply		This pin is Positive Source driving voltage - VSH1.	VSH1R
VSH2	P	VSH2 supply		This pin is Positive Source driving voltage - VSH2.	VSH2R
VSL	P	VSL supply		This pin is Negative Source driving voltage.	VSLR

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